

DYNAMIC STOPPING ENVELOPE: LINEAR REACTION LAG VS. QUADRATIC BRAKING

Stopping Distance $d_{stop} = v \cdot T_{lag} + v^2 / (2 \cdot a_{max})$ across Forward Speed v and Actuator Deceleration Capacity a_{max}

Phase 1: Linear Reaction Travel (Lag-Bound)

$$d_{react} = v_0 \cdot T_{lag} = v_0 \cdot (t_{sense} + t_{inf} + t_{bus} + t_{elec})$$

Constant-velocity unguided displacement while policy evaluates

At $v = 2.0 \text{ m/s}$ and $T_{lag} = 120 \text{ ms}$: $d_{react} = 24.0 \text{ cm}$ (Linear scaling: $\propto v$)

Linear $\propto v$

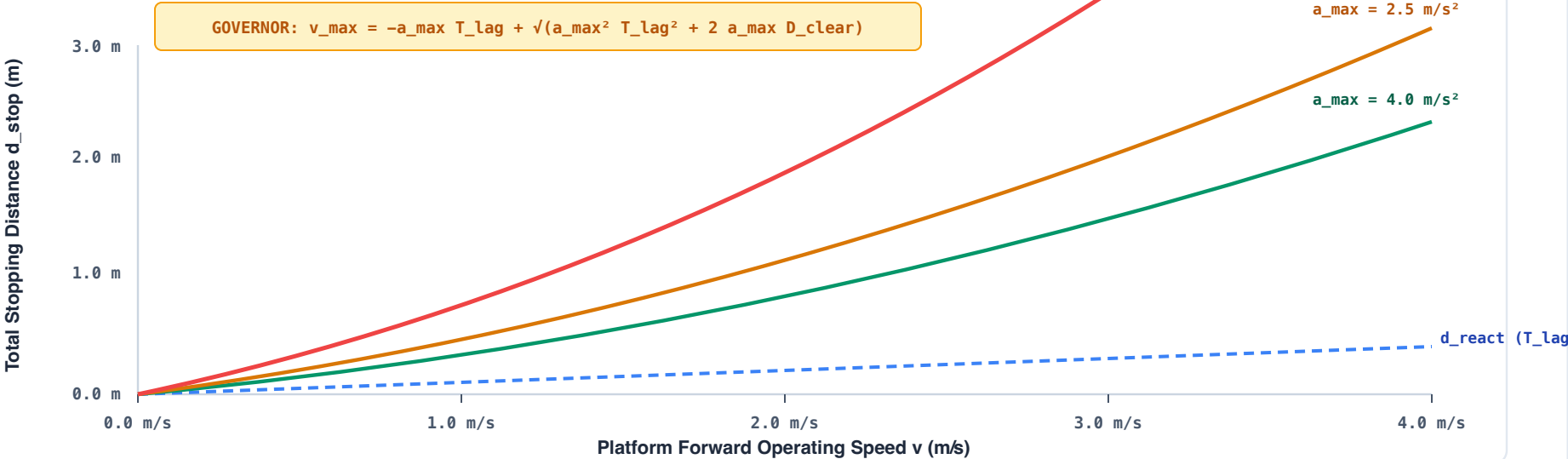
Phase 2: Quadratic Braking Travel (Physics-Bound)

$$d_{brake} = v_0^2 / (2 \cdot a_{max}) \cdot (1 + k_{fudge})$$

Kinetic energy dissipation: $\Delta E_k = \frac{1}{2} m v^2$ converted to Joule heat

At $v = 2.0 \text{ m/s}$ and $a_{max} = 3.0 \text{ m/s}^2$: $d_{brake} = 66.7 \text{ cm}$ (Quadratic: $\propto v^2$)

$a_{max} = 1.5 \text{ m/s}^2$ (Wet / Low Friction) Quadratic $\propto v^2$



As speed increases, quadratic kinetic braking dominates total stopping distance, requiring non-linear velocity clamps as obstacle clearance shrinks.